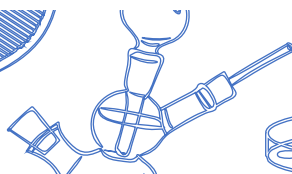
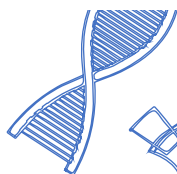
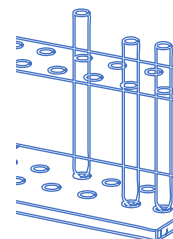
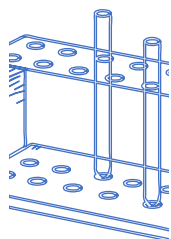


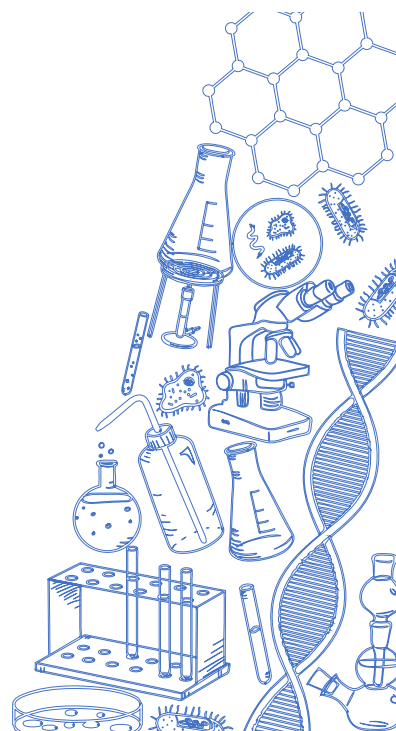
Science G10

Chemical Reactions Class 3



Agenda

1. Take up Class 1 Homework
2. Quiz 3
3. Lesson 3
 - Arrhenius Acids and Bases
 - Neutralization Reactions
 - Acid-Base Indicators
 - pH Scale
 - Applications of pH
 - Acid Deposition

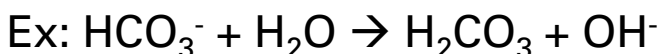
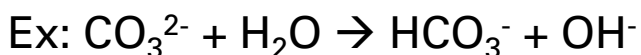


Arrhenius Acid and Base

- **Arrhenius acid** is a substance that produces H^+ when dissolved in water



- **Arrhenius base** is a substance that produces OH^- when dissolved in water



Svante Arrhenius (1859-1927)

Properties of Acids and Bases

	ACIDS	BASES	
 	• Conducts electricity • Tastes sour • Turns blue litmus red • Neutralizes base • Commonly found in: • Preservatives • Stomach • Citrus fruits • Soda	• Conducts electricity • Tastes bitter • Turns red litmus blue • Neutralizes acid • Commonly found in: • Cleaning agents • Antacids • Baking soda	 

Names of Acids

Common Binary Acids

Chemical Formula	Acid Name
HF (aq)	Hydrofluoric acid
HCl (aq)	Hydrochloric acid
HBr (aq)	Hydrobromic acid
HI (aq)	Hydroiodic acid
H ₂ S (aq)	Hydrosulfuric acid

Common Oxyacids

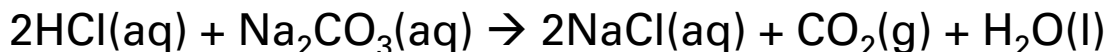
Chemical Formula	Acid Name
CH ₃ COOH (aq)	Acetic acid
H ₂ CO ₃ (aq)	Carbonic acid
HNO ₃ (aq)	Nitric acid
H ₃ PO ₄ (aq)	Phosphoric acid
H ₂ SO ₄ (aq)	Sulfuric acid

Reaction of Acids

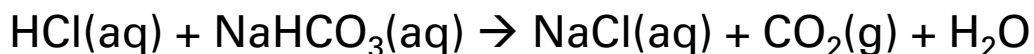
- **Acid with Metals → Hydrogen Gas**



- **Acid with Carbonates → CO₂**

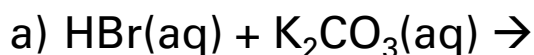


- **Acid with Hydrogen Carbonates → CO₂**



Checkpoint

Predict the products of the following reactions and balance:



Names of Bases

Common Bases

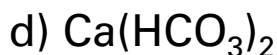
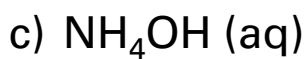
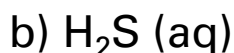
Chemical Formula	Acid Name
NaOH (aq)	Sodium hydroxide (aka: lye or caustic soda)
LiOH (aq)	Lithium hydroxide
KOH (aq)	Potassium hydroxide (aka: caustic potash)
Ca(OH)_2	Calcium hydroxide (aka: slaked lime)
Mg(OH)_2	Magnesium hydroxide (aka: milk of magnesia)
Al(OH)_3	Aluminum hydroxide
NaHCO_3	Sodium hydrogen carbonate (aka: baking soda)



Checkpoint

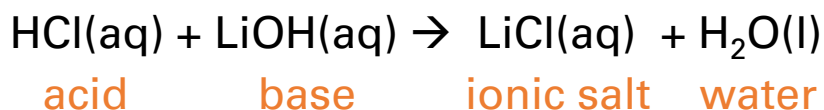


Name the following and categorize each as an acid or base:



Neutralization Reactions

- **Neutralization reactions** is the reaction of an acid and base to form an ionic salt and water
 - A subcategory of double displacement reactions

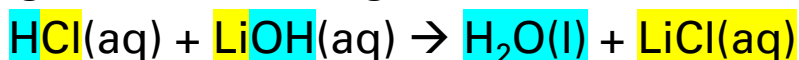


Predicting the Products of Neutralization Reactions

1. Highlight the H^+ from the acid and OH^- from the base to form $\text{H}_2\text{O}(\text{l})$



2. Highlight the remaining ions to form the ionic salt.



3. Balance the chemical reaction.



Checkpoint

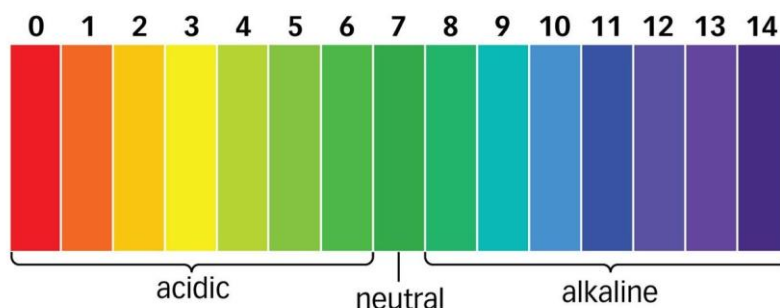


Predict the products of the following neutralization reactions and balance:

- a) Hydrobromic acid + Sodium hydroxide
- b) $\text{HI}(\text{aq}) + \text{NH}_4\text{OH}(\text{aq})$
- c) Phosphoric acid and Barium hydroxide

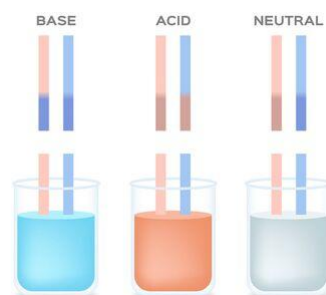
Acid-Base Indicators

- **Acid-base indicator** is a substance that changes colour when added to an acid or base



Indicator	Colour in Acid	Colour in Base
Phenolphthalein*	Colourless	Pink
Litmus*	Red	Blue
Methyl orange	Red	Orange/Yellow
Bromothymol blue	Yellow	Blue
Phenol red	Yellow	Red/Pink

*Memorize for G10 Science



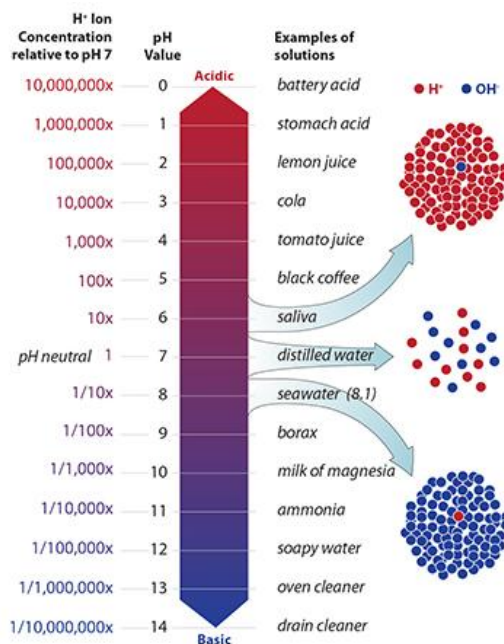
Phenolphthalein 8,2–10,0			
Methyl Orange 3,0–4,4			
Bromthymol Blue 6,0–7,6			
Phenol Red 6,8–8,4			

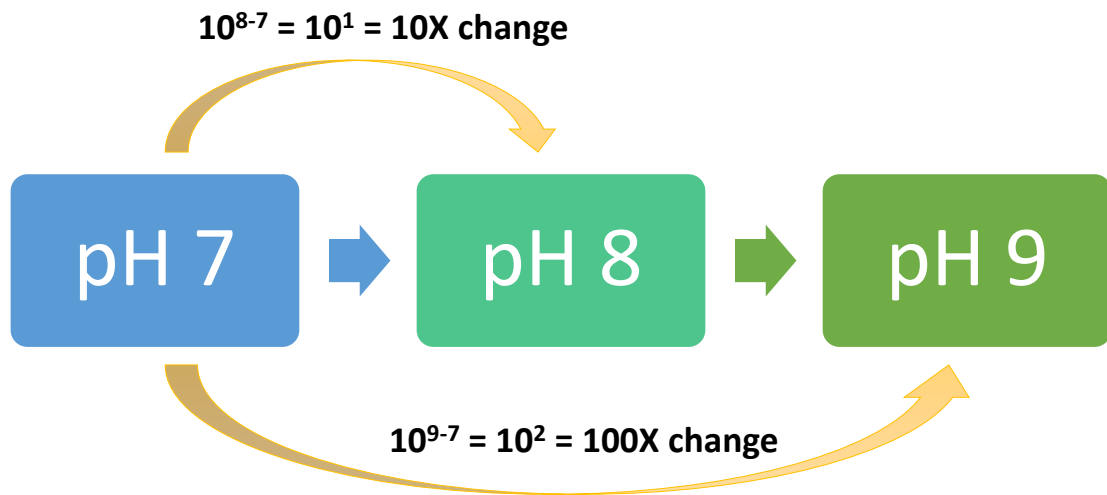
STOP Checkpoint STOP

You are looking for the identity of Substance X. If you add phenolphthalein, Substance X turns pink. Is Substance X acidic or basic?

pH Scale

- **pH** stands for the “power of Hydrogen” and is a measure of acidity
- pH scale uses a logarithmic scale
- At 25°C:
 - $\text{pH} < 7$ is acidic
 - $\text{pH} = 7$ is neutral
 - $\text{pH} > 7$ is basic
- Practical pH range is 0-14





Checkpoint

List the following from most acidic to least acidic:

Substance	pH
Milk	6
Lemon Juice	2
Vinegar	3
Ammonia	11

Checkpoint

Substance A has a pH of 13.2 Substance B has a pH of 8.5.

- Which substance is more acidic?
- Which substance is more basic?
- How many times did the concentration change from Substance A to Substance B?

pH in Agriculture

- Plants are sensitive to soil pH
 - Legumes grow well in slightly basic soil (pH 7-10)
 - Corn thrives in mildly acidic soil (pH 5-6)
 - Potatoes grow best in acidic conditions (pH <5)



Colours of hydrangeas depend on soil pH

Acid Leaching

- **Acid leaching** is a process of adding acid to soil that has been contaminated with heavy metals that are basic
- **Phytoremediation** is the process of using plants to absorb metal toxins
 - Sunflowers absorb toxic metals and radioactive elements
 - Aspen trees remove lead from water



pH in Consumer Products

- Cleaning products are very acidic or basic and can be corrosive to skin and membranes

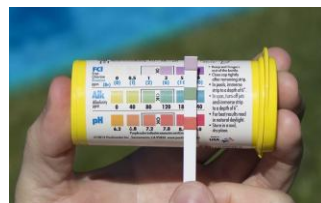
Product	pH
Bleach	11-13
Oven Cleaner	11-13
Windex	9
Dish Soap	7-8
Toilet Bowl Cleaner	1-3
Battery Acid	0-1

- Wear rubber gloves and ensure area is well-ventilated

WHMIS 2015 symbol
for Corrosive material



- pH of swimming pools should be 7.2-7.8
 - If pH is less than 7, pool water will irritate eyes
 - If pH is greater than 8, pool water becomes cloudy and chlorine compound will lose its disinfecting ability
- Pool pH test kits are used to monitor pH
 - Add HCl(aq) (muriatic acid) to reduce pool pH
 - Add $\text{Na}_2\text{CO}_3\text{(aq)}$ to increase pool pH

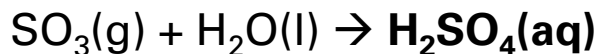
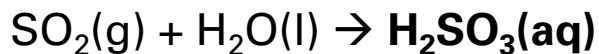


Acid Deposition

- Rainwater is naturally acidic ($\text{pH} = 5.6$) due to the presence of dissolved CO_2 to form $\text{H}_2\text{CO}_3\text{(aq)}$
- **Acid Deposition (pH below 5.6)** is the mix of air pollutants that lead to acidification of soil and water systems
 - **Wet Acid Deposition** comprises of acid rain, snow, sleet, hail, fog, mist, or dew
 - **Dry Acid Deposition** comprises of gases that fall to the ground as dust or smoke which later dissolve in water bodies to form acids

Main Contributors of Acid Deposition

- **Sulfur oxides (SO_x)** produced from the burning of fossil fuels



- **Nitrogen oxides (NO_x)** produced from the burning of fossil fuels

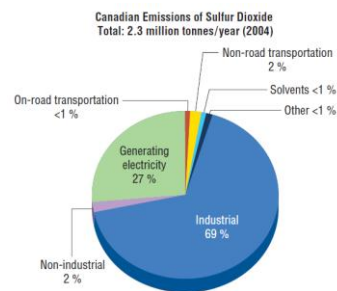


Figure 2 Sources of sulfur dioxide emissions in Canada

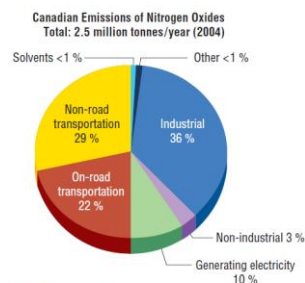
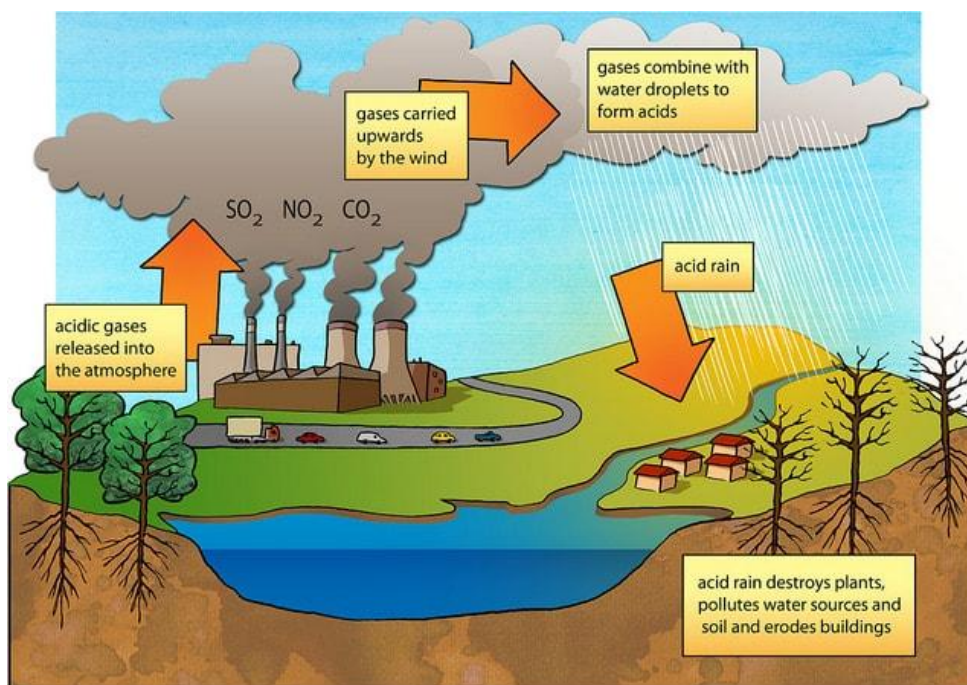


Figure 4 Sources of nitrogen oxide emissions in Canada



Effects of Acid Deposition

- Damages forests and plants
 - Acid rain cause slower growth, injury to leaves, blocks stomata for gas exchange
 - Leaches minerals from the soil
- Reduction in aquatic life
 - Acid release toxic Al^{3+} from rocks which interferes with gill function
 - Nitrogen compounds lead to algal blooms and eutrophication
- Damages buildings and structures



What I Learned Today:

- ☐ Arrhenius Acids and Bases
- ☐ Neutralization Reactions
- ☐ Acid-Base Indicators
- ☐ pH Scale and Applications
- ☐ Acid Deposition

Due next class: Class 3 Homework

